Quantum Optimization for Fast CAN Bus Intrusion Detection

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ABSTRACT

Modern vehicles, nowadays, come loaded with hundreds of different sensors generating a huge amount of data. This data is shared and processed among different Electronic Control Units (ECUs) through an in-vehicle network, such as the CAN bus, to improve the driver's experience and safety. However, the implementation of new features increases exposure to cyber-attacks. The CAN bus, which is designed to grant reliable communication, has many security weaknesses that might be exploited by an attacker. The need for highly accurate real-time intrusion detection systems (IDSs) for the automotive industry is limited to classical machine learning techniques, which are usually time-consuming and have hardware limitations. In this work, we analyze an optimized and efficient version of a network-based IDS for CAN bus attack detection based on Quantum Annealing. The models were tested on two different CAN bus datasets. The results show that the Quantum Annealing algorithm outperforms a classical classification algorithm in terms of time performance, which is important in the identification of attacks in the automotive sector, and achieves similar results for detection accuracy.

CCS CONCEPTS

• **Software and its engineering → Real-time systems software**; • **Security and privacy → Intrusion detection systems**; • **Computing methodologies → Machine learning algorithms**.

KEYWORDS

in-vehicle security, quantum computing, quantum annealing, cybersecurity

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1 INTRODUCTION

Almost all modern vehicles come with an onboard computer. These computers interact with numerous sensors across the vehicles to create a safe and sound driving experience for the driver. These new generation vehicles are characterized by ECUs (Electronic Control Units) which are components aimed at making decisions based on sensor detection. These components communicate with each other using various protocols such as Ethernet, FlexRay, Local Interconnect Network (LIN), Media Oriented Systems Transport (MOST), Controller Area Network (CAN) [9] etc. One of the most used protocols is CAN for its safety and velocity to send messages. However, CAN is vulnerable to different types of attacks such as Denial of Service (DoS), eavesdropping, injection, and replay attacks [7]. Since a successful attack might have serious impacts on passengers' safety due to anomalous vehicle behaviour, cybersecurity plays an extremely important role in this critical field. Different solutions have been proposed to overcome these security weaknesses, such as network segmentation, encryption, authentication and IDSs [5]. Many of the traditional machine learning-based solutions are usually time-consuming and don't meet real-time requirements which are fundamental in the automotive field. On the other hand, lightweight algorithms are faster but less accurate [4]. The main challenge is to detect attacks on the CAN bus in shortest time without affecting the accuracy of the algorithm negatively, emphasizing the need for new real-time solutions. The interest in quantum computing has largely grown in the last few years. Apart from the new threats and different attack scenarios that quantum computing will introduce, it could also be seen as an opportunity and a powerful means to strengthen the security of the systems already in use. Especially in critical scenarios, such as the automotive industry, where fast detection of anomalies are extremely important for an early response. In this research work, we propose an optimized and efficient version of a CAN bus intrusion detection system with quantum annealing. The main objective of this work is to investigate how a quantum classification algorithm can outperform a classical classification algorithm. We test our system using 2 different attack signatures performed on the CAN bus: DoS and Fuzzy. Each dataset contains more than 3.5 million samples and a small number of features. The rest of the paper is organized as follows: Section 2 presents some of the important related works; The CAN protocol is then briefly introduced in Section 3; Empirical evaluation of the proposed system is discussed in Section 4 with elaborations on methodology, datasets and experimentation, and Section 5 presents the conclusion and future works.

2 RELATED WORKS

Several works have been proposed in the literature that aim to detect attacks on the Controlled Area Network (CAN) protocol. The authors have analyzed different existing approaches in [8], defining a taxonomy of problems where machine learning is currently used in cybersecurity. They analyzed three main areas where cyber ML algorithms are being used, namely intrusion detection, malware analysis and spam detection. The authors concluded that although the ML algorithms are promising for cyber security, they are still vulnerable to adversarial attacks. Cho and Shin proposed an efficient lightweight voltage-based attacker identification system, called Viden, that is able to determine which Electronic Control Unit (ECU) is responsible for the attack by measuring its output voltage on the CAN bus [6]. In another approach [3], the authors proposed an efficient intrusion detection system to identify attack messages sent on a Controller Area Network (CAN) bus, integrating an unsupervised Kohonen Self-Organizing Map (SOM) network with the K-Means algorithm. Barletta et al. proposed Artificial Intelligence (AI) techniques in other application, for example, IoT [16]. In literature, there aren't many works focused on the application of quantum computing in automotive cybersecurity. In [1], the researchers proposed a Quantum Optimization for IoT security detection. In [14], the authors focused on two different quantum approaches, Quantum Swarm Evolutionary (QSE) algorithm and the Quantum-inspired Evolutionary Algorithm (QEA) for detecting DoS attacks on cloud computing. Our method aims to define a new IDS capable of identifying attacks on the CAN bus in the shortest possible time. In particular, in the experimental section, considering two datasets, we evaluated a quantum classification algorithm against a traditional classification algorithm from the viewpoint of time performance.

3 BACKGROUND

3.1 Controller Area Network

The CAN bus is the most used protocol in-vehicle network. The CAN protocol was released for the first time in 1986. However, it became a standard for on-board and off-board communications in 1993 when the International Organization for Standardization (ISO) published the ISO 11898 for road vehicles. A typical frame used for the attack phase is *CAN data frame*, and it contains various fields, namely, an *Arbitration field* (11 bits for CAN 2.0A or 29 bits for CAN 2.0B) carrying the frame ID, a *Control field* (1 bit) for the control signal, which also contains the data length, and a Data field (64 bits) containing the actual data.

3.2 QBoost in CAN Security Detection

In classical machine learning, ensemble algorithms combine predictions of multiple base estimators to increase system performance, leading to better results when there is high diversity among each and every single estimator. QBoost was proposed in [11], and is derived from one of the classical ensemble learning techniques and is specifically designed from quantum annealing, whose main goal is to optimize the best combination of learners for a given set of training data by minimizing the quadratic deviation between observed and predicted labels. It introduces a penalty for complex models that include many weak classifiers [12]. In the automotive environment identifying an attack as early as possible could prevent unexpected behavior in vehicles and even save the driver's life. For this reason, fast and quick response times for attack identification are crucial, especially for CAN, as it is the most used protocol in the in-vehicle network.

4 EMPIRICAL EVALUATION

4.1 Methodology

In order to prove the effectiveness and optimization capabilities of quantum annealing in detecting CAN bus attacks, the performance of a hybrid quantum-classical classifier was compared to that of a classical one. For the experiment phase, we compared two binary classifiers: QBoost algorithm and Support Vector Machine (SVM) using SVC as its implementation from Scikit-learn[1]. Google and D-Wave [13] have already described how the QBoost ensemble algorithm could be used efficiently to extend binary classification to quantum computing. The use of SVM is discussed by Mukkamala et al. [10] in which they have reached good results for DoS attacks. Tanksale [19] used SVM to detect anomalous behavior with high accuracy in the automotive field. Thus, its SVC implementation from Scikit-learn was used for this purpose here. We choose two types of datasets: DoS and Fuzzy because they have two different structures and different types of injection attacks. Since previous works proved that classifiers trained to detect a specific type of attack perform better than generic ones, the two binary classifiers, SVC and QBoost were trained for each dataset. The datasets were split into 80% training and 20% testing set before training. In order to reduce bias and balance the ratio of CanIds between the two splits [2] [15], every dataset was shuffled. We used the generic IDS pipeline (Figure 1) during the experiments for both SVC and QBoost, which is summarized below:

- *Pre-processing & Feature extraction*: Removal of messages with data field shorter than 8 bytes ('DLC' != 8) since classifiers can't handle missing values. The *Frequency* feature was derived from the timestamp, representing the time in milliseconds between two successive messages on the CAN bus with the same CanId. To handle categorical data, two types of encoding techniques: One-Hot-Encoding and binary encoding on *CanId* and string *Data* obtained by merging data bytes Data[0-7]. Finally, each feature was individually scaled to the range [min, max] with MinMax scaler. Labels used for classification were also mapped from 'R': non-malicious, 'T': malicious to '+1': non-malicious, '-1': malicious.
- *Feature Selection*: For each experiment a subset of N features, consisting of 'Frequency' and attributes derived from encoding 'CanId' and 'Data', were selected.
- *Model Training*: For each dataset two binary classifiers, SVC and QBoost were trained.
- *Prediction*: Test data was provided as input to the two trained models separately to make predictions.

We used the Leap Quantum Cloud Service[2] that provides access to a portfolio of hybrid solvers located in different geographical

[1] https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html
[2] https://www.dwavesys.com/solutions-and-products/cloud-platform/

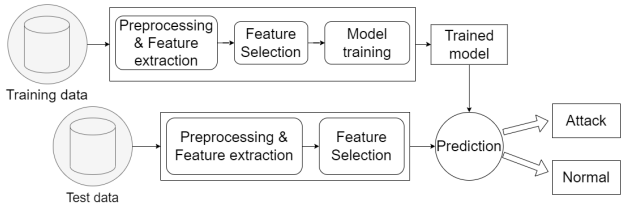


Figure 1: IDS pipeline

regions. A back-end server consists of Quantum Processing Units (QPUs) and Solvers (a resource that runs a problem by directly interfacing to the QPU or by leveraging its CPU and GPU) which accept jobs and return the solution of the problem. We carry out the QBoost experiments using D-Wave.

4.2 Dataset

We analyze the classifier performances using Car-Hacking Dataset for intrusion detection that were constructed by logging CAN traffic via the OBD-II port from a real vehicle while message injection attacks were performing [17, 18]. For this work we used two datasets: DoS and Fuzzy. A DoS attack consists of injection messages with dominant arbitration id (0000). The messages were injected every 0.3 milliseconds. This type of attack could flood the CAN bus because in the arbitration phase the dominant bit has a higher priority as compared to the recessive bit (1-bit). Instead, in a Fuzzy attack, spoofed random CAN ID and Data values are injected every 0.5 milliseconds that causes unusual behaviour of the vehicle.

The structure of the messages present in the dataset consist of the following 5 attributes:

- *Timestamp*: the time (in seconds) the message was recorded;
- *CanId*: identifier of a message in HEX;
- *DLC*: the number of data bytes from 0 to 8 that was sent on the CAN bus;
- *DATA[0-7]*: byte value expressed in HEX;
- *Label*: T or R, T represents injected message while R represents normal message.

4.3 Experimentation

4.3.1 DoS Experiments: Data in DoS dataset has a simple structure formed by 25 unique CanId messages and malicious messages sent with CanId '0000' and null Data payload. We applied One-Hot-Encoding[3] on *CanId* and binary encoding[4] on *Data*. The one-hot-encoding technique creates a new binary variable for each unique category, therefore extracting N features out of N categories, where 0 and 1 indicate the absence or presence of that category in each row. The total number of features were 43 including *Frequency*. We computed Accuracy, Precision, Recall and F1-score to compare the classification performance. The results of two binary classifiers are shown in Table 3. Both classifiers were able to distinguish between malicious and non-malicious messages, reaching 100% Accuracy, Precision, Recall, and F1-score in each of the 10 rounds of testing. As data shuffling before every round didn't affect the metrics, and

[3] https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html
[4] https://contrib.scikit-learn.org/category_encoders/binary.html

Table 1: Average time performance of the Binary classifiers on DoS dataset. Grey shows best performance

Classifier	Average training time (s)	Average prediction time (s)
SVC	68.6735	2.5609
QBoost	26.1415	0.0275
	-42.532	-2.5334

Table 2: Average time performance of the Binary classifiers on Fuzzy dataset. Gray shows best performance

Classifier	Average training time (s)	Average training time (h)	Average prediction time (s)	Average prediction time (m)
SVC	5933.2840	1.6481	431.4609	7.191
QBoost	25.7134	0.0071	0.1462	0.0024
	- 5907.5705	-1.641	- 431.3146	-7.1886

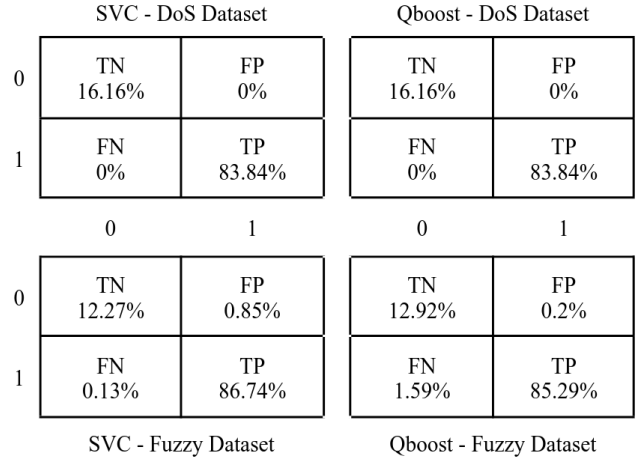


Figure 2: Confusion matrix for SVC and QBoost on DoS & Fuzzy datasets. "0" malicious; "1" normal messages. TN-True Negative, FP-False Positive,FN-False Negative,TP-True Positive

the training and testing set metrics were the always the same, we might safely assume these to be good indicators of low overfitting. In fact, the confusion matrices of SVC and QBoost in Figure 2 shows that each attack (label 0) and normal message (label 1) in the testing set was correctly classified. The confusion matrices shown in Figure 2 refers to the results from the last run of the test set.

QBoost showed a considerable improvement in the execution times. Table 1 compares the performance in seconds between the two classifiers. We can see that we have a high gain in execution time in both the training and prediction phases, with gains of 62% and 99%, respectively.

4.3.2 Fuzzy Experiments: The Fuzzy database has a more complex structure compared to the DoS, since malicious messages are randomly sent to the CAN bus using all the 2017 unique CanIds, 37 of which also transmits normal traffic. Binary encoding was applied to both *CanId* and *Data* which returned 11 and 20 features

Table 3: Classification performance of binary classifiers: Train & Test sets for DoS & Fuzzy dataset. Gray shows best performance

Dataset	Set	Classifier	Accuracy	Precision	Recall	F1-score
DoS	Train	SVC	1.0	1.0	1.0	1.0
		QBoost	1.0	1.0	1.0	1.0
	Test	SVC	1.0	1.0	1.0	1.0
		QBoost	1.0	1.0	1.0	1.0
Fuzzy	Train	SVC	0.9858	0.9945	0.9891	0.9918
		QBoost	0.9778	0.9988	0.9757	0.9871
	Test	SVC	0.9716	0.9785	0.9892	0.9838
		QBoost	0.9756	0.9959	0.9760	0.9858

respectively, making 32 features in total with *Frequency*. The results of two binary classifiers are shown in Table 3. Considering the recall metric, we can see that the SVC performed slightly better than QBoost. From the viewpoint of time performance, Table 2 shows the average results obtained from 10 runs and it can be seen that QBoost clearly outperformed SVC in terms of execution time, during both training and prediction phases, with 99% gain in both cases. The confusion matrices for SVC and QBoost shown in Figure 2 confirms the precision and recall metrics presented in Table 3. For the fuzzy attack, the confusion matrices also refer to the last run of the testing set. The training time for SVC was closer to that of QBoost for DoS dataset results. On the other hand, the Fuzzy set being more complex in comparison to DoS, the acquired results prove that quantum annealing based algorithm successfully managed to optimize a complex problem.

5 CONCLUSION & FUTURE DIRECTION

In this work, we have used two open-source datasets, DoS and Fuzzy, based on CAN bus attacks. We have compared the classification as well as time performance between QBoost and SVC. We saw that QBoost improves the training and testing times remarkably as compared to SVC. The time improvement was also obtained with the Fuzzy dataset which has a complex structure compared to the DoS dataset. The threat of fuzzing attacks on a in-vehicle CAN bus system is very much real. As the proposed model helps in extremely fast CAN bus intrusion detection, it might be genuinely helpful for real time attack identification. The proposed model shows an extensive improvement in time performance while maintaining similar quality for CAN bus attack classification. The analysis of other classification algorithms both classical and quantum, e.g. Random Forest or AdaBoost, and experimentation on other public datasets can be considered in future works for further investigation.

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